TEJAS TOVINKERE PATTABHI

Staff Engineer · Cloud · Distributed Systems · Applied AI

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QUALIFICATIONS FOR THE U.S. ASSIGNMENT

- **Advanced degree in the field.** M.S. Computer Science, The University of Texas at Dallas (08/15/2015), on top of a four-year B.E. in Information Science (First Class with Distinction); Ph.D. in Artificial Intelligence in progress.
- **10+ years of U.S. software engineering experience** building large-scale distributed systems, cloud infrastructure and applied AI, at Qualcomm (09/27/2021 to present) and Webroot/OpenText (09/14/2015 to 09/24/2021).
- **Specialized automotive and AI expertise.** Lead engineer for the infrastructure that validates Qualcomm's automated-driving software (Snapdragon Ride), key collaborator on the Qualcomm–BMW Group program, and project lead of a production AI agent that triages tens of thousands of pipeline failures a day.
- **Named co-inventor on two granted U.S. patents** (US 11,201,875 B2 and US 12,452,259 B2) and first author of a peer-reviewed publication; invited peer reviewer for IEEE GSCon 2026.

CORE SKILLS

Languages	Python · Java · C# · C · Bash	**Cloud**	AWS · Google Cloud (Dataflow) · Azure
Orchestration	Kubernetes · Argo Workflows · Argo Events · Docker · Helm	**Applied AI**	LLM agents (Claude Agent SDK · AWS Bedrock) · agentic coding (Claude Code · Codex) · custom Claude skills & plugins · MCP tool integration
ML & vectors	Self-hosted embedding models (Qwen3, ONNX int8) · pgvector cosine similarity · semantic deduplication	**IaC & Ops**	AWS CDK · Prometheus · Grafana · CI/CD · IPMI/BMC hardware mgmt
Architecture	Distributed & event-driven systems · petabyte-scale data pipelines	**AWS services**	ECS Fargate · SQS · EventBridge · RDS/PostgreSQL · S3 · AppSync/GraphQL · IAM
Databases	PostgreSQL/pgvector · MySQL · MS-SQL		

PROFESSIONAL EXPERIENCE

Qualcomm Technologies, Inc.

09/27/2021 – Present

5775 Morehouse Drive, San Diego, CA 92121, USA · Full time

Staff Engineer	11/25/2023 – Present
Senior Engineer	09/27/2021 – 11/24/2023

Autonomous Driving Data Development team, Qualcomm's ADAS/autonomous-driving program. Builds the large-scale infrastructure that replays recorded real-world driving data through automated-driving software to validate perception, radar, vehicle self-motion/positioning, ground-truth labeling and automatic emergency braking (AEB) before the software reaches a vehicle.

RESPONSIBILITIES

- Design, develop and maintain large-scale distributed software that reprocesses recorded drive data and computes validation KPIs for automated-driving functions.
- Develop and operate orchestration software for on-premises Kubernetes compute clusters built on automotive accelerator hardware: job scheduling, dependency resolution, event-driven workflows and parallel transfer of petabyte-scale data.
- Integrate new accelerator hardware into the clusters, give feedback on the board support package, deploy it at scale and expose it to engineering teams as a self-service processing service.
- Develop a cloud and on-premises reprocessing platform behind a single interface (AWS, Google Cloud Dataflow), including input selection, failure classification and retry logic.
- Lead the design and development of a production AI system that triages pipeline failures, diagnoses root cause and proposes fixes.

- Build monitoring and maintenance software for the compute fleet: health and firmware monitoring, dashboards, IPMI/BMC power control and safe node maintenance.
- Define infrastructure as code (AWS CDK); maintain CI/CD pipelines, automated testing and observability.
- Set technical direction, review code daily and coordinate engineering across teams in three time zones; mentor and onboard engineers and interns; interview engineering candidates.

PROJECTS

- **Automotive-grade reprocessing clusters.** As a lead engineer, contributed extensively to building and operating two independent on-premises Kubernetes clusters of Qualcomm **QCR-100** cards (the same Snapdragon Ride silicon that ships in vehicles), dedicated to ADAS/AD reprocessing, so validation runs on the exact silicon the software executes on in the car rather than an approximation from generic hardware.
- **Reprocessing-as-a-Service platform.** Architected a pipeline that runs identically in the cloud or on-premises: job orchestration that selects input files (AppSync/GraphQL + IAM role assumption), resolves inter-step dependencies into a DAG, and submits event-driven **Argo Workflows** (Argo Events); plus parallel data transfer, input population, cleanup, and retryable/infrastructure/application failure classification. Assisted in integrating the QCR-100 into custom hosted cloud environments so partners and customers can reach this specialized compute on demand.
- **Qualcomm–BMW Group program.** Key collaborator on the reprocessing pipeline behind the jointly developed Snapdragon Ride Pilot automated-driving system, turning recorded driving data into the validated results behind the platform.
- **Production AI pipeline-triage agent.** As project lead, built and run an agent that triages job failures across cloud and on-premises pipelines. A deduplication service embeds each failure (self-hosted embedding model) and uses **pgvector cosine-similarity** search to suppress recurring failures, so only novel ones reach a **Claude Agent SDK + AWS Bedrock** agent that gathers diagnostics, classifies root cause, files Jira tickets and proposes GitHub configuration-fix pull requests, escalating ambiguous cases to engineers. Event-driven, cross-account AWS (ECS Fargate, SQS FIFO, EventBridge Pipes, RDS/pgvector, Managed Grafana); AWS CDK with full CI/CD and observability.
- **Fleet-health and throughput monitoring.** Built a Prometheus + Flask monitoring stack: a per-node DaemonSet that reports each accelerator's health and firmware every 30 seconds and tracks **Real-Time Factor (RTF)**, the ratio of processing time to the duration of the recorded drive, with a live dashboard of available compute and throughput.
- **Hardware-maintenance tooling.** Built the cluster maintenance layer: IPMI/BMC server power control, firmware-flash (EDL) mode control, and safe Kubernetes node drain/cordon and uncordon, so hardware can be serviced without disrupting running jobs.
- **Massively parallel cloud reprocessing (Google Cloud Dataflow).** Re-architected the cloud reprocessing pipeline onto Dataflow, redesigning how work is partitioned and scheduled to raise concurrency and cut end-to-end turnaround while improving output accuracy.

ACCOMPLISHMENTS

- Contributed to the validation pipeline behind the Qualcomm–BMW Snapdragon Ride Pilot system, publicly launched in September 2025 in the BMW iX3 and validated for use in 60 countries.
- Purpose-built silicon clusters run reprocessing about **100–300× faster than general-purpose compute**, fast enough to replay recorded drives at close to real time and meet customer-delivery timelines.
- AI triage agent handles **tens of thousands of pipeline failures per day**, so engineers see only novel failures, with root cause and proposed fixes attached.
- Promoted from Senior Engineer to Staff Engineer effective 11/25/2023.

Open Text, Inc. (formerly Carbonite Inc. and Webroot Inc.)

09/14/2015 – 09/24/2021

Work location: 4655 Executive Drive, Suite 500, San Diego, CA 92121, USA · Full time (at least 40 hours per week)

Open Text, Inc.	01/01/2021 – 09/24/2021	Sr. Software Engineer
Carbonite Inc.	01/01/2020 – 12/31/2020	Sr. Software Engineer
Webroot Inc.	09/14/2015 – 12/31/2019	Software Engineer, then Sr. Software Engineer

Carbonite acquired Webroot on 03/26/2019 and OpenText acquired Carbonite on 12/24/2019; employment continued without a break through each transfer. Internet-scale cybersecurity and threat-intelligence systems.

RESPONSIBILITIES

- Develop and maintain back-end software that enhances the security of applications using Webroot's threat-intelligence platforms.
- Contribute to new feature designs, build new software features, and develop systems to monitor uptime and accuracy of software products.

- Develop and maintain compute clusters for multi-level web analysis and threat categorization.
- Develop and maintain analytical rules to enhance the IP threat database; implement and maintain IP systems to identify and act on malicious IPs.
- Architect and deploy honeypots on multiple cloud platforms to trap malware, attackers and proxies.
- Build these systems on cloud computing platforms (AWS, Azure, Google Cloud); own AWS hosting and DevOps.
- Provide on-call production support for one week in every four.

PROJECTS

- **Web crawling, classification and threat detection at internet scale.** Led distributed crawling clusters that crawled and classified threats across a **peak of 17M+ URLs/day** in real time to protect end users.
- **DeepCrawl.** Co-invented a technique that pivots from a single known malicious link to uncover related malicious infrastructure and grades its threat level with a machine-learning model.
- **IP threat analysis.** Subject-matter expert and lead; built a system that categorized malicious hosts by activity (phishing, botnet, malware, proxy, spam) to publish threat lists.
- **Cloud security intelligence and reputation.** Designed a reputation-scoring system for cloud applications and their hosting organizations, gathering data vectors dynamically to surface security posture to customers.
- **Multi-cloud honeypots.** Architected honeypots across AWS, Google Cloud and Azure that captured live attacks (scanners, malware, injections, DoS) to continuously feed the threat-analysis systems.

ACCOMPLISHMENTS

- DeepCrawl became the basis of **two granted U.S. patents** (named co-inventor): US 11,201,875 B2 (12/14/2021) and US 12,452,259 B2 (10/21/2025), assignee Open Text Inc.

TradeStation Technologies, Inc.

05/16/2014 – 12/31/2014

2600 N Central Expy, Richardson, TX 75080, USA

Software Developer Intern

- Built an AWS Elastic MapReduce (EMR) application that analyzed daily IIS web server logs.
- Visualized request and response volumes with D3.js.

Societe Generale Global Solution Centre Private Limited

08/08/2012 – 07/26/2013

Voyager Bldg, International Tech Park Bangalore, Whitefield Road, Bangalore, Karnataka 560066, India

Young Graduate, Windows Server Administrator

- Administered production Windows Server environments across VMware ESXi hosts in two data centers.
- Automated production-server health checks and presented server status as a web application.

EDUCATION

Ph.D. Artificial Intelligence, University of the Cumberlands, Williamsburg, KY · In progress (started Fall 2026).

M.S. Computer Science, The University of Texas at Dallas, Richardson, TX · Fall 2013 to Summer 2015 · Degree conferred 08/15/2015.

B.E. Information Science and Engineering, PES School of Engineering, Bangalore (Visvesvaraya Technological University), India · 2008 to 2012 · Completed 07/25/2012 · First Class with Distinction in all eight semesters.

ACADEMIC PROJECT

Award Solutions, Richardson, TX: Student Volunteer Intern, OpenStack (05/27/2015 – 08/05/2015). Built a webhook-driven application that monitored VM state transitions in Ceilometer; the project work was published as open source by UT Dallas.

PATENTS, PUBLICATIONS AND PEER REVIEW

U.S. Patent 12,452,259 B2 (10/21/2025) and **U.S. Patent 11,201,875 B2** (12/14/2021): "*Web Threat Investigation Using Advanced Web Crawling,*" named co-inventor, assignee Open Text Inc.

Publication (first author): "Implementing Delaunay Triangles and Bezier Curves to Identify Suitable Business Locations in the Presence of Obstacles," *International Journal of Information Technology and Computer Science (IJITCS)*, vol. 5 no. 3, 2013.

Reviewer, IEEE Gujarat Section Conference (GSCon), 2026: reviewed two submitted papers on invitation.